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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Powerstrangers
Selected Domain	Domain 1 – Future-Ready Workforce Domain
Selected SDG Track (1–17)	SDG 4
Project / Solution Name	FutureReady MY
Name (Leader)	Mak Jia Hng
Email & Phone (Leader)	ivanjiahngmak@gmail.com (+60189708879)
University Affiliation(s)	University of Malaya, University Tunku Abdul Rahman, Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia's technology job seekers and career changers often make upskilling decisions using fragmented information. Generic career advice global trend reports, and larger course catalogues do not clearly show which skills are evidenced in Malaysian job postings for a specific technology role. This creates uncertainty for students, new graduates, and professionals transitioning into roles such as Software Engineer, Data Engineer, Data Scientist, Machine Learning or AI Engineer.

The challenge is not simply compared with locally observed employer requirements, identify the gaps that matter most, and find relevant learning resources without relying on unsupported claims or black-box recommendations. Without this guidance, learners may spend time and money on courses that do not address their most relevant skill gaps.

This mismatch can contribute to difficulties in preparing talent for Malaysia's digital economy. It also affects equitable access to career guidance, since people without professional networks may have fewer ways to interpret labour-market expectations.

FutureReady MY addresses this problem by translating processed Malaysian job-posting evidence into explainable skill-demand insights, role-readiness comparisons, and traceable learning pathways. The platform supports SDG 4 by aligning technical upskilling with employment-relevant skills, while also providing documented links to SDG 8 and SDG 9. It does not claim to guarantee employment or measure direct SDG impact.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FutureReady MY is an AI-assisted workforce-intelligence platform that connects Malaysian job-market evidence to personalized career actions. Users can upload a PDF resume or enter their skills and experience. Before any AI processing, the system redacts sensitive information such as Malaysian NRIC numbers, phone numbers, email addresses, and street address. Qwen2.5-7B-Instrucct, served through a RunPod Serverless vLLM workers, can extract structured skills from unstructured text. A verification gate then accepts only canonical skills with evidence in the user's input, if the model fails or violates guardrails, the system uses a deterministic dictionary fallback. The platform analyses 63,852 deduplicated Malaysian JobStreet postings to show role-specific skills, posting shares, and evidence limits. Role matching is explainable and deterministic. It combines empirical skill coverage (70%) with TF-IDF similarity (30%), and is explicitly not a probability of being hired. For missing skills, FutureReady MY retrieves traceable course from an 8370-record Coursera dataset using hybrid semantic and lexical search, then explain the relevant skill gap and heuristic learning order. The system also presents documented, non-measured alignment to SDGs 4, 8, and 9.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React 19, TypeScript, Vite, Tailwind CSS

Backend: Python 3.11+, FastAPI, Pydantic and Uvicorn

Data: JobStreet Malaysia postings; Coursera course catalog; IT-skills taxonomy; processed Parquet/CSV metadata

AI: Qwen2.5-7B-Insturct through RunPod Serverless vLLM; PII redaction, skill verification, factual-fidelity checks, and deterministic fallbacks

Retrieval: all-MiniLM-L6-v2 CPU ONNX embeddings + TF-IDF reciprocal-rank fusion

Persistence: SQLite for local development, with PostgreSQL-compatible migrations

Deployment: RunPod CPU FastAPI pod on 8000; RunPod Network Volume for datasets and indexes

Architecture flow: User -> React web app -> FastAPI API -> (JobStreet/Coursera processed data, role matcher, course retriever) -> optional RunPod vLLM worker -> explainable results and learning pathway

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our design-thinking process begins by empathising with Malaysian students, job seekers, and career changers who need clearer evidence for technology-career decisions. We define the problem as an information gap between scattered learning choices and locally observed employer skill requirement. We ideate around one connected journey, understand a profile, explore Malaysian demand, compare suitable roles, and follow a transparent learning pathway. The prototype combines a responsive React and TypeScript interface with a FastAPI intelligence service. We will test core flows using representative profiles invalid input, sparse role samples, and unavailable AI-service scenarios. the service is designed for RunPod deployment. A CPU FastAPI pod handles data processing and retrieval, while a scale-to-zero Serverless vLLM worker is used only when AI extraction or verified narration is enabled. Privacy controls, deterministic fallbacks, explicit source coverage, and non-hiring-guarantee language keep the prototype usable and responsible during Phase 2.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4, Target 4.4, increase the number of youth and adults with relevant technical and vocational skills for employment.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	0 completed FutureReady MY user pathways measured. the prototype has processed 63,852 Malaysian postings and identified 1553 postings across its five supported role families.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage of identified core or recommended skill gaps for which the system returns at least one traceable course recommendation, plus the number of tested user profiles receiving an explainable role-match result.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 90% verified pathway coverage for test profiles with supported roles and identifiable skill gaps. report the numerator, denominator, unresolved gaps, and limitations.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [ ] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (Code/Dataset/ Model)	License
JobStreet Malaysia postings WEF report dataset	Dataset	Kaggle Open Data (CC BY 4.0)
Coursera Course dataset	Dataset	CC BY-NC 4.0 (Non-commercial)
IT Skill taxonomy	Dataset	Apache 2.0
WEF Report	Dataset	Public Report
Qwen2.5-7B-Instruct	Model	Qwen Research License (Apache 2.0)

sentence-transformers	Model	Apache 2.0
all-MiniLM-L6-v2	Model	Apache 2.0
FastAPI	Code	MIT
React	Code	MIT
TypeScript	Code	Apache 2.0
Vite	Code	MIT
Tailwind CSS	Code	MIT
ONNX Runtime	Code	MIT
SQLite	Code	Public Domain

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature



Date: 24/10/2026